

JNTU Kakinada

MID-TERM EXAMINATION ? 2026

Theory of Computation ? Mid-Term Examination, 2026

Subject Code: CS504 Semester: 5 Regulation: R23

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Finite Automata)

1. Solve a problem related to Finite Automata. Show all the steps clearly. [12 Marks]
2. Write short notes on Finite Automata. [8 Marks]
3. Compare and contrast the key concepts involved in Finite Automata. [8 Marks]

SECTION B (Unit 2: Regular Expressions and Languages)

4. List the advantages and limitations of Regular Expressions and Languages. Explain any two in detail. [12 Marks]
5. How is Regular Expressions and Languages implemented? Give an algorithm / procedure and explain. [12 Marks]
6. List the advantages and limitations of Regular Expressions and Languages. Explain any two in detail. [12 Marks]

SECTION C (Unit 3: Context Free Grammars)

7. Describe the working principle of Context Free Grammars with necessary equations. [12 Marks]
8. Define Context Free Grammars. Explain its significance in Theory of Computation with a suitable example. [8 Marks]
9. Compare and contrast the key concepts involved in Context Free Grammars. [12 Marks]

SECTION D (Unit 4: Pushdown Automata)

10. Describe the working principle of Pushdown Automata with necessary equations. [12 Marks]
11. List the advantages and limitations of Pushdown Automata. Explain any two in detail. [12 Marks]
12. List the advantages and limitations of Pushdown Automata. Explain any two in detail. [12 Marks]

SECTION E (Unit 5: Turing Machines and Complexity)

13. Write short notes on Turing Machines and Complexity. [12 Marks]
14. Define Turing Machines and Complexity. Explain its significance in Theory of Computation with a suitable example. [8 Marks]
15. Discuss the engineering applications of Turing Machines and Complexity. [8 Marks]

*** End of Question Paper ***